Unit 4 – Cloud Security Assignment

Topic: Launch a Windows VM and Connect Using RDP

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1. Aim

To launch a Windows Server virtual machine on AWS EC2, allow RDP access only from my own IP address, connect to the VM using Remote Desktop Protocol (RDP), create a new user account with a strong password, verify the Windows Firewall, and create a text file on the server.

2. Requirements

- An active AWS account (Free Tier eligible t2.micro instance).
- A key pair (.pem file). For Windows instances it is used to decrypt the Administrator password.
- A Windows PC with the Remote Desktop Connection app (mstsc), or an RDP client on another OS.
- The public IP address of my own computer (for the Security Group rule).

3. Procedure

Note: As permitted in the assignment instructions, the screenshots in this report are AI-generated illustrative images. They are labelled as such below.

Step 1: Launch a Windows Server VM on AWS EC2

- Log in to the AWS Management Console, select the region Asia Pacific (Mumbai) and open EC2.
- Click "Launch instance" and enter the name "Windows-Server".
- Select the AMI: Microsoft Windows Server 2022 Base (Free tier eligible).
- Select the instance type: t2.micro (1 vCPU, 1 GiB memory).
- Select an existing key pair or create a new one (RSA, .pem) and download it.
- In Network settings, keep the default VPC and enable "Auto-assign public IP".
- Create a new security group named "windows-rdp-sg" (see Step 2).
- Click "Launch instance" and wait until the Instance state is "Running" and the status checks show "2/2 checks passed".

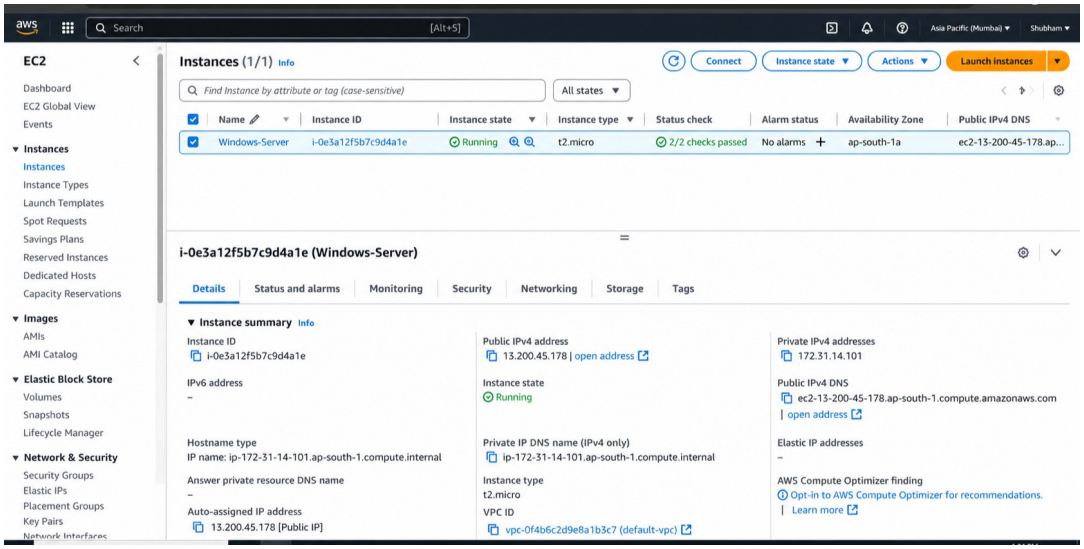


Figure 1: EC2 instance Windows-Server in Running state (with public IPv4 address) (AI-generated illustrative image)

Step 2: Configure the Security Group for RDP

The Security Group acts as a virtual firewall for the VM. One inbound rule was added:

Type	Protocol	Port	Source	Purpose
RDP	TCP	3389	My IP (single IP, /32)	Remote desktop login to the Windows server

RDP is allowed only from my own public IP address. It is not opened to 0.0.0.0/0, because an RDP port open to the whole internet is a common target of brute-force attacks. The default outbound rule (all traffic) was kept.

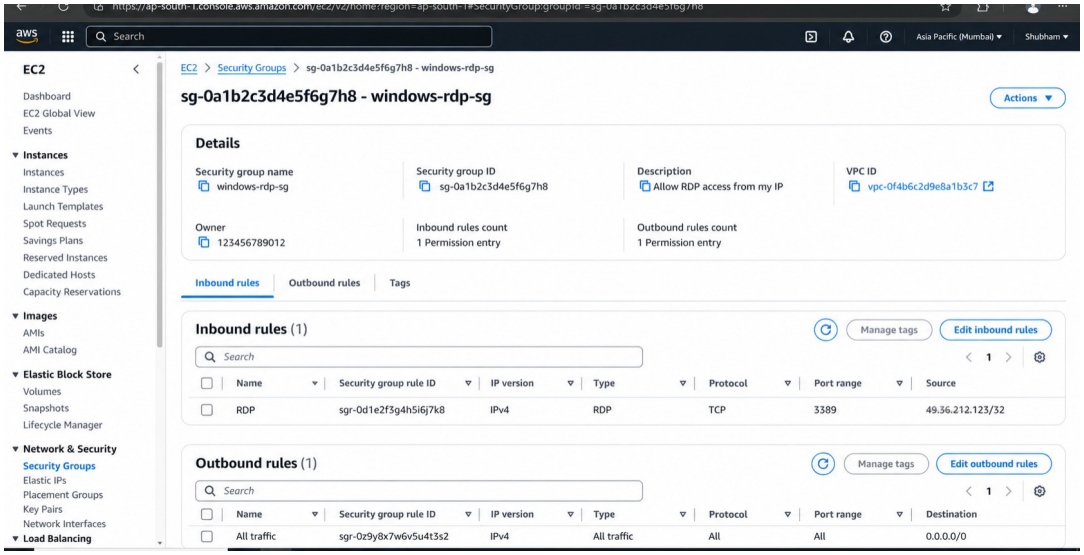


Figure 2: Security Group windows-rdp-sg – inbound rule RDP (TCP 3389) from my IP (AI-generated illustrative image)

Step 3: Connect to the Windows VM using RDP

- In the EC2 console select the instance and click Connect, then open the "RDP client" tab.
- Click "Get password", upload the .pem key file and click "Decrypt password". Copy the Administrator password (it is not written in this report).
- Open Remote Desktop Connection (press Win + R, type mstsc, press Enter).
- In Computer enter the public IPv4 address of the instance, click Connect.
- Username: Administrator, Password: the decrypted password. Accept the certificate warning.

Connection details (password excluded):

Item	**Value**
Operating system	Windows Server 2022 Standard
Instance type	t2.micro
Region	Asia Pacific (Mumbai) – ap-south-1
Public IPv4 address	13.200.45.178 (see Figure 1)
Protocol / Port	RDP / TCP 3389
Username	Administrator
Password	Not shown (kept secret)

Step 4: Explore the Windows Server desktop and check system information

After logging in, Server Manager opens automatically. System information was checked by pressing Win + R, typing msinfo32 and pressing Enter (or by running "systeminfo" in Command Prompt).

```
msinfo32
systeminfo
```

The System Summary shows the OS name (Microsoft Windows Server 2022 Standard), system manufacturer (Amazon EC2), system model (t2.micro), processor, installed memory (1 GB) and the time zone.

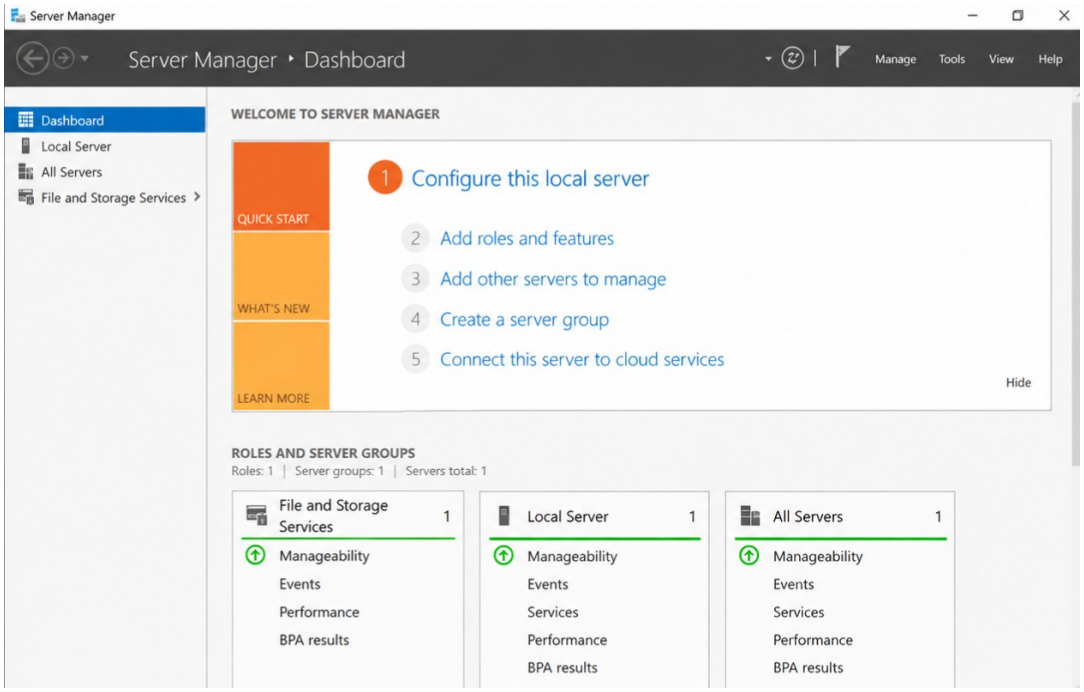


Figure 3: Server Manager dashboard (AI-generated illustrative image)

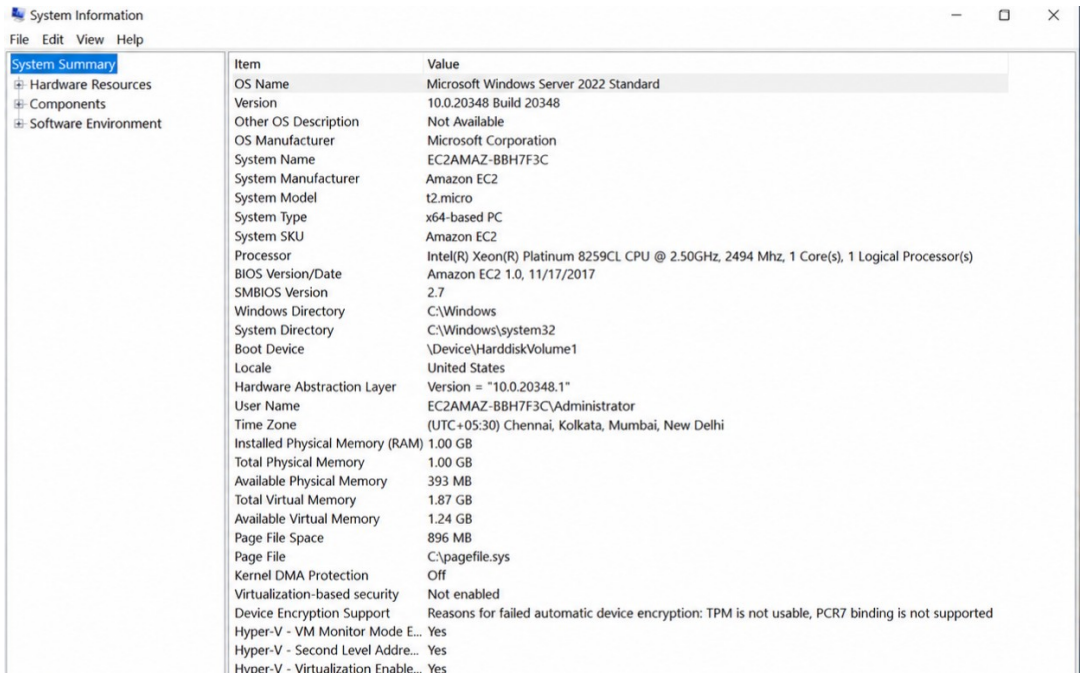


Figure 4: System Information (msinfo32) – OS, system model and memory (AI-generated illustrative image)

Step 5: Create a new user account with a strong password

A standard user account was created so that daily work is not done with the Administrator account. Open Computer Management (Win + R, type compmgmt.msc) and go to System Tools > Local Users and Groups > Users. Right-click and select "New User". The same can be done from Command Prompt (run as Administrator):

```
net user shubham * /add
net localgroup "Remote Desktop Users" shubham /add
```

The "*" makes Windows ask for the password without showing it on the screen. The user name is shubham, full name Shubham Paithankar and description "User account for assignment". A strong password was used:

- At least 12 characters long.
- A mix of uppercase and lowercase letters, numbers and special characters.
- No name, PRN, date of birth or dictionary word.
- Not reused from any other account (the password is not written in this report).

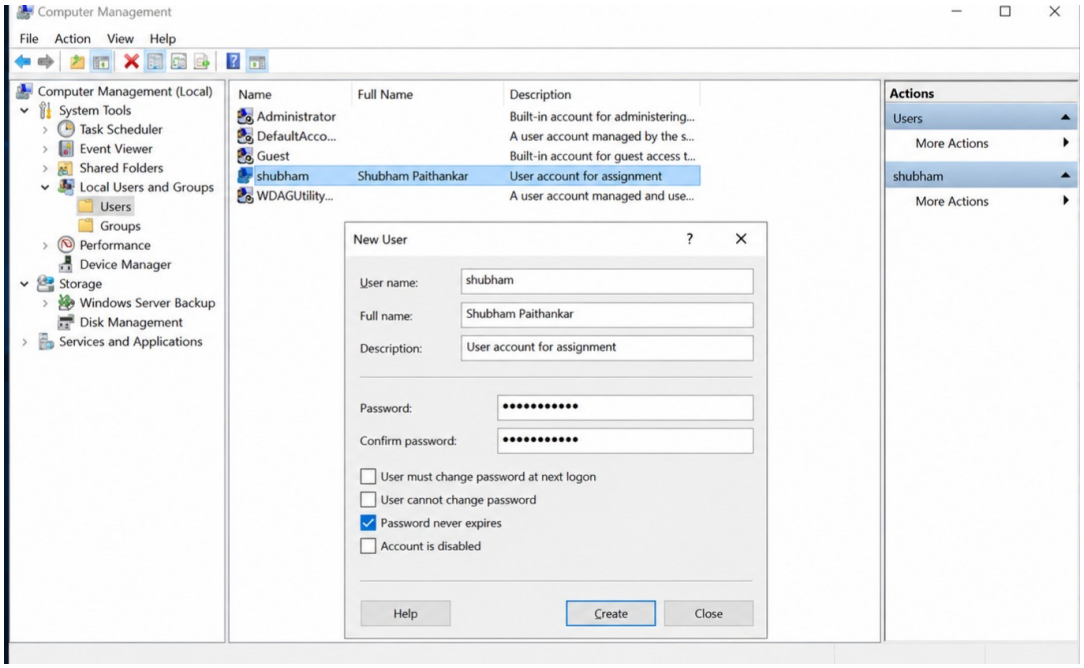


Figure 5: Computer Management – new user account created (AI-generated illustrative image)

Step 6: Enable Windows Firewall and verify its status

The firewall status was checked from Control Panel > System and Security > Windows Defender Firewall. It can also be enabled and verified using Command Prompt:

```
netsh advfirewall set allprofiles state on
netsh advfirewall show allprofiles
```

Windows Defender Firewall state should be "On" for the Domain, Private and Public profiles, with incoming connections blocked for apps that are not on the allowed list.

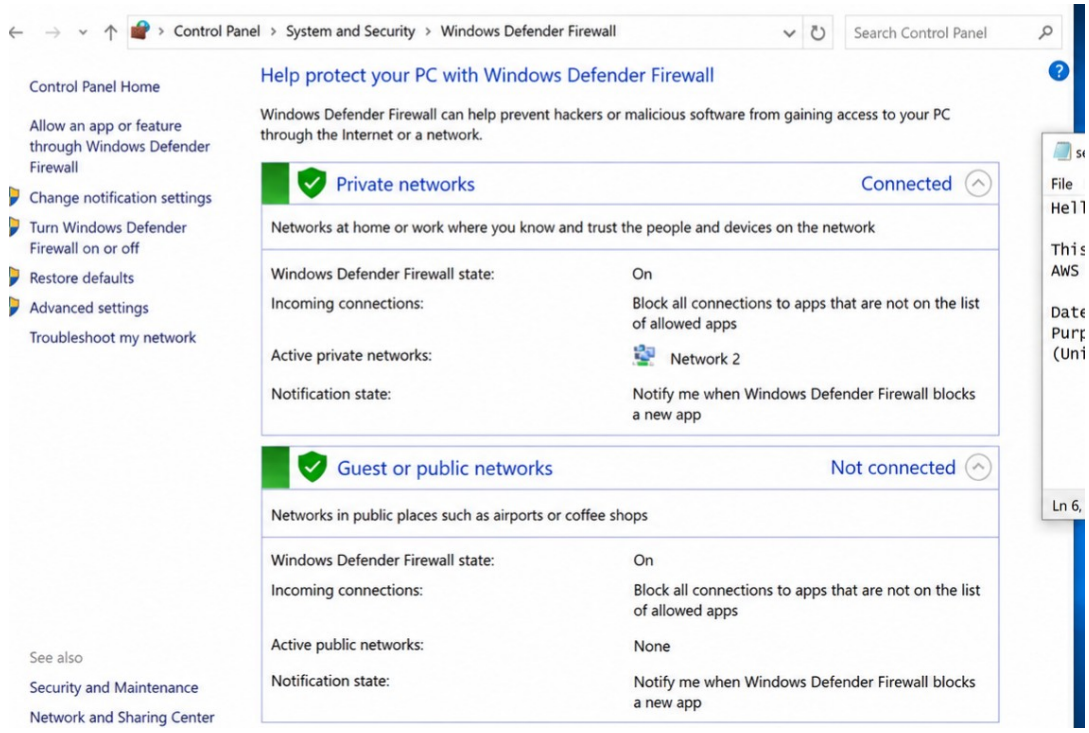


Figure 6: Windows Defender Firewall – state On for private and public networks (AI-generated illustrative image)

Step 7: Create a text file on the server

A text file was created to confirm that the server can be used. Right-click on the Desktop, select New > Text Document, name it (for example server-test.txt), open it in Notepad, type a short message with the name, PRN, date and purpose, and save it.

4. Security Configuration Summary

Area	Configuration
Network access	RDP (TCP 3389) allowed only from my own IP (/32); not open to 0.0.0.0/0
Authentication	Administrator password decrypted with the private key file; new standard user created
Passwords	Strong password policy followed; passwords not shared or written in the report
Host firewall	Windows Defender Firewall turned On for all network profiles
Access control	Day-to-day work done with a standard user instead of Administrator
Key file	The .pem file is stored privately and never shared

Additional good practices: keep Windows updated, stop or terminate the instance when it is not in use to avoid charges, and use a VPN or AWS Systems Manager Session Manager instead of exposing RDP publicly in a real environment.

5. Result

A Windows Server 2022 virtual machine was launched on AWS EC2, the Security Group was configured to allow RDP only from my IP address, and the VM was accessed using Remote Desktop Protocol. System information was checked, a new user with a strong password was created, the Windows Firewall was verified to be on, and a text file was created on the server.

6. Conclusion

This assignment showed how a Windows virtual machine can be launched and managed in the cloud using RDP. It highlighted the importance of restricting RDP access to a trusted IP address, using strong passwords, using a separate user account instead of Administrator, and keeping the host firewall enabled to reduce the risk of unauthorized access.